



## RB004 Week 4 — English Worksheet

**Topic:** RemoteEvents — Arena Announcements · **Course:** Arena Legends · **Time:** about 45 minutes

This week the server learns to talk to players. A **RemoteEvent** in ReplicatedStorage is like a megaphone: the server **shouts** with `FireAllClients`, and every player's screen **listens** with `OnClientEvent` in a LocalScript.

### 1 · Predict 🧐

Read each script. Before you imagine running it in Studio, write what you think will happen.

```
-- Script in ServerScriptService
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local announceEvent = ReplicatedStorage.AnnounceEvent

announceEvent:FireAllClients("Round started!")
```

You test with 2 Players. The server fires the event. How many players receive the message "Round started!"?

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```
-- LocalScript inside the TextLabel
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local announceEvent = ReplicatedStorage.AnnounceEvent
local label = script.Parent

announceEvent.OnClientEvent:Connect(function(message)
    label.Text = message
end)
```

The server fires `FireAllClients("Round started!")` . What does this label show? What does it show on the OTHER player's screen?

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```
-- Script in ServerScriptService
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local Players = game:GetService("Players")
local announceEvent = ReplicatedStorage.AnnounceEvent

local winner = Players.Player1
announceEvent:FireClient(winner, "You win!")
```

`FireClient` sends to ONE player. You test with 2 Players. What does Player1's screen show? What does Player2's screen show?

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## 2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

**Bug A** — Every player's screen should show the server's announcement. But this listener never runs, because LocalScripts cannot run in ServerScriptService.

```
-- LocalScript in ServerScriptService
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local announceEvent = ReplicatedStorage.AnnounceEvent
local label = game.StarterGui.ArenaGui.AnnounceLabel

announceEvent.OnClientEvent:Connect(function(message)
    label.Text = message
end)
```

**Hint:** the listener belongs on the **player's side** — for example, a LocalScript inside the TextLabel in StarterGui.

**Write the fixed code (including where it goes):**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The server should announce "Round started!" to **every** player. Instead, the script errors and nobody sees anything.

```
-- Script in ServerScriptService
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local announceEvent = ReplicatedStorage.AnnounceEvent

announceEvent:FireClient("Round started!")
```

**Hint:** `FireClient` needs a **player** first — but here we want to shout to everyone, so a different function fits better.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — The player's screen should listen for the server's message. But the listener uses the wrong ear and never runs.

```
-- LocalScript inside the TextLabel
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local announceEvent = ReplicatedStorage.AnnounceEvent
local label = script.Parent

announceEvent.OnServerEvent:Connect(function(message)
    label.Text = message
end)
```

**Hint:** the **client** listens with `OnClientEvent`. `OnServerEvent` is for the server's ear, not the player's.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Tell Me What You Built 📷

Now open your own arena game. Add a RemoteEvent named **AnnounceEvent** in ReplicatedStorage, make the server fire "Round started!" when the round begins, and make a TextLabel show the message on every player's screen. Test with **Test tab** → **2 Players**. When you finish, come back here.

Send a photo or video of the announcement appearing on both players' screens, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I put the RemoteEvent in **ReplicatedStorage** because ...

The server shouts with **FireAllClients** when ...

The player's screen listens with **OnClientEvent** and then ...

One tricky moment was when ...

If I had more time, I would ...

## 4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you test your announcements with 2 Players. Talk through it like you are teaching someone who has never seen it. Try to use these words:

**RemoteEvent, server, client, FireAllClients, OnClientEvent.**

1. Show the RemoteEvent in ReplicatedStorage and your two scripts — say which one is the server and which one is the client.
2. Press Test → 2 Players and show the announcement appearing on both screens.
3. Read the `FireAllClients` line and the `OnClientEvent` line out loud, and say which one shouts and which one listens.
4. Say in your own words why the server cannot change a player's screen directly.

**Write what you will say in your video.** Use the space below to plan it before you record — you can read from it while filming.

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